Hyung Cho

AI Engineer | Python GenAI Applications | Retrieval, Grounding, Tool Calling, and LLM Evaluation

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SUMMARY

AI engineer who builds Python GenAI applications with retrieval and grounding, tool calling, and the evaluations that gate them, using AI coding agents such as Claude Code as the core workflow. Before that, 10 years of product validation at Zebra Technologies, ending with two years building C# test automation.

EXPERIENCE

Independent AI Engineer | Self-directed projects, Remote **Jun 2025 - Present**

Solo product work, built by directing AI coding agents (Claude Code, Codex).

Purvia, permission-aware RAG and support-bot app, open source

- Built a multi-user RAG chat app on FastAPI and the raw OpenAI SDK with multi-format document ingestion, Pydantic structured-output metadata extraction, streamed responses, and per-chunk source badges in a React and TypeScript UI.
- Implemented hybrid retrieval that fuses pgvector HNSW search with Postgres full-text search by reciprocal rank fusion, with a keyword fallback and optional rerankers; a bake-off kept reranking off by default.
- Built a framework-free tool-calling agent loop, capped at 5 iterations, in which the model chooses between document search, text-to-SQL, web search, and a full-document sub-agent that runs in its own isolated context.
- Enforced access control inside the Postgres retrieval functions so every search runs as the signed-in user, backed by row-level security and a CI eval that fails unless a user with no grants gets nothing protected back.
- Built a 60-question retrieval regression set scored on recall@k, MRR, and nDCG@5 that posts the PR-versus-main difference on pull requests; hybrid search scores 0.950 recall@5.
- Designed an escalation eval whose LLM judges come from two providers, OpenAI and Anthropic, with a weekly 5% false-resolve ceiling and a judge-agreement check; it has flagged a breach four weeks running.
- Tagged LangSmith traces by user and thread, and set up scheduled eval jobs that commit dated results and open an issue when a run fails or measures nothing.
- Made model choice configurable per role across OpenAI, Azure OpenAI, and OpenAI-compatible endpoints, and deployed a demo with a Dockerized FastAPI backend on Railway and the frontend on Vercel.

Rep Today, native iOS workout app, pre-release

- Built a premium coaching chat that sends a user's message and a non-identifying training summary through a Cloudflare Worker proxy to OpenAI, with App Attest and subscription checks on the server, capped output, and 462 proxy tests in CI.
- Directed and approved 196 pull requests written by AI coding agents, 180 of them with a written risk assessment.

Volunteer AI/ML Engineer | Omdena, Remote **Apr 2023 - Jun 2025**

Unpaid collaborative ML projects with global volunteer teams.

- Fine-tuned open-source LLMs (LLaMA, Falcon) and iterated on chatbot prompts with an evaluation workflow.

Career break and ML retraining **Aug 2018 - Mar 2023**

Self-directed ML study and the FourthBrain ML Engineering bootcamp (2022); 29 public ML projects on GitHub.

Senior Validation Engineer | Zebra Technologies, Agoura Hills, CA **Feb 2008 - Jul 2018**

- Built C# test automation that replaced manual workflows and cut regression cycles by about 30%.
- Led product validation for 6 shipped hardware products: test plans, procedures, acceptance criteria, and final reports.
- Isolated root causes of failures across hardware, firmware, software, and test infrastructure, and proved the fixes.

SKILLS

GenAI: RAG, pgvector, Prompt engineering, Tool calling, Text-to-SQL, OpenAI API, Azure OpenAI, Anthropic API
LLMOps: LLM evaluation, LLM-as-judge, RAGAS, Retrieval metrics, LangSmith, GitHub Actions, CI/CD, Docker
Engineering: Python, FastAPI, Pydantic, PostgreSQL, SQL, Supabase, TypeScript, React, Cloudflare Workers, C#
AI developer tools: Claude Code, AI coding agents, Code review, Architecture decision records

EDUCATION AND TRAINING

B.S., Electrical Engineering, California Polytechnic State University, San Luis Obispo
FourthBrain ML Engineering bootcamp (2022); DeepLearning.AI deep learning and generative AI courses